

MCP Server Setup Guide

Steps to Configure and Run the JMeter MCP Server on Another Machine

Share your GitHub repository with your friends and ask them to follow the below steps carefully.

1. Clone the GitHub Repository

Open terminal or CMD.

Run:

```
</> Bash
git clone <YOUR_GITHUB_REPO_URL>
```

Example:

```
</> Bash
git clone https://github.com/username/jmeter-mcp-server.git
```

2. Open Project in VS Code

Open VS Code.

Go to:

```
File → Open Folder
```

Select cloned project folder.

Expected structure:

```
jmeter-mcp-server/
|
├─ node_modules/
├─ tests/
|   └─ sample-test.jmx
|
├─ jmeter.log
├─ jmeter.properties
└─ package-lock.json
```

```
├─ package.json
├─ README.md
├─ result.jtl
├─ server.js
└─ server2.js
```

3. Install Node.js

Install Node.js from:

[Node.js Official Website ↗](#)

Verify installation:

```
</> Bash
```

```
node -v
```

```
npm -v
```

4. Install Apache JMeter

Download JMeter from:

[Apache JMeter Official Website ↗](#)

Extract it.

Example path:

```
C:\Jmeter\apache-jmeter-5.6.3\apache-jmeter-5.6.3
```

5. Verify JMeter Installation

Open terminal inside JMeter `bin` folder.

Run:

```
</> Bash
```

```
jmeter -v
```

If version appears:

✅ JMeter installed correctly.

6. Open Project Terminal

Inside VS Code terminal run:

```
</> Bash  
npm install
```

This installs all required dependencies from `package.json`.

7. Update JMeter Path in server.js

Open:

```
server.js
```

Find:

```
</> JavaScript  
const jmeterPath =  
  "C:/Jmeter/apache-jmeter-5.6.3/apache-jmeter-5.6.3/bin/jmeter.bat";
```

Update this path according to local JMeter installation path.

Example:

```
</> JavaScript  
const jmeterPath =  
  "D:/Tools/apache-jmeter-5.6.3/bin/jmeter.bat";
```

Important:

- ✓ Use forward slashes `/`
- ✗ Do NOT use backslashes `\`

8. Verify jmeter.properties Exists

Open:

```
jmeter.properties
```

Ensure it contains:

```
</> properties
```

```
threads=100
```

```
rampup=10
```

```
loops=5
```

9. Start MCP Server

Open terminal inside project folder.

Run:

```
</> Bash
```

```
npm start
```

Expected output:

```
JMeter MCP Server Running...
```

Keep this terminal running.

10. Install VS Code Extensions

Inside VS Code Extensions install:

- GitHub Copilot
- GitHub Copilot Chat

Optional but recommended for MCP integration.

11. Configure MCP Server in VS Code

Press:

```
Ctrl + Shift + P
```

Search:

```
MCP: Add Server
```

Choose:

Command (stdio)

12. Enter MCP Configuration

Command

node

Arguments

Use full path of `server.js`.

Example:

C:/Users/HP/Desktop/jmeter-mcp-server/server.js

13. Restart VS Code

Close VS Code completely.

Reopen VS Code.

14. Verify MCP Server Connection

Check VS Code output logs.

Expected:

Starting server my-mcp-server
Discovered 1 tools

This confirms:

- ✓ MCP connected
- ✓ Tool registered successfully

15. Verify MCP Tool

Press:

Ctrl + Shift + P

Search:

MCP: Show Tools

Expected tool:

execute_and_optimize_jmeter_test

16. Run JMeter Test Through MCP

Use prompt:

Use the execute_and_optimize_jmeter_test MCP tool to execute and optimize this JMeter

C:/Users/HP/Desktop/jmeter-mcp-server/tests/sample-test.jmx

Update path according to local machine.

17. Expected Workflow

AI Prompt

↓

MCP Server

↓

Run JMeter

↓

Generate result.jtl

↓

Analyze Results

↓

Detect Bottlenecks

↓

Update jmeter.properties

↓

Return Optimization Summary

18. Expected Output Example

JMeter Test Completed Successfully

Total Requests: 500

Failed Requests: 20

Average Response Time: 1800 ms

Error Rate: 4%

BOTTLENECK DETECTED:

System stable under current load.

OPTIMIZATION APPLIED:

Increased threads to 120 for next stress level.

UPDATED CONFIGURATION:

Threads: 120

Ramp-up: 10

19. Files Automatically Generated

File	Purpose
result.jtl	JMeter execution results
jmeter.log	JMeter logs
jmeter.properties	Auto-updated optimization config

20. Important Notes

Keep MCP Server Running

Do NOT close terminal running:

```
</> Bash
```

```
npm start
```

Always Use Full Paths

Correct:

```
C:/Users/HP/Desktop/jmeter-mcp-server/tests/sample-test.jmx
```

Wrong:

```
sample-test.jmx
```

Use Forward Slashes

Correct:

```
C:/Jmeter/apache-jmeter/bin/jmeter.bat
```

Wrong:

```
C:\Jmeter\apache-jmeter\bin\jmeter.bat
```

Final Outcome

This MCP server project enables:

- ✓ Natural language JMeter execution
- ✓ Automatic result analysis
- ✓ Bottleneck detection
- ✓ Automatic optimization of `jmeter.properties`
- ✓ AI-generated performance summaries

using MCP and Apache JMeter integration.